

Intelligent Shared Control Prosthetic System with Tactile Enhancement & Skill Transfer

Taoxiang Fang, Junfeng Wang, Yichen Jie, Lingyue Yang, Chenyu Zhang, Yuxin Bi, Zihao Lv, Cheng Peng, Yunzhao Fu, Yaxu Li, Yifei Liu.

Project Supervisor: Dr Bin Fang, Ms Pei Qie, Dr Liang He.

1. Introduction

Intelligent prosthetics is a frontier field integrating bionic robotics, AI, and rehabilitation engineering. Current products lack effective tactile perception, stable motion intent decoding, and adaptive human-machine shared control, leading to low efficiency, severe muscle fatigue, and limited dynamic operation capability.

Zhiyu Lingzhi is an intelligent human-machine shared prosthetic system with tactile enhancement and skill transfer. It combines multi-modal tactile sensing, myoelectric decoding (DiffSR model), collaborative control, and dynamic skill transfer (TKE- BGC model) to achieve autonomous force control, low-fatigue manipulation, and human-like tool operation.

Backed by a complete technical architecture and independent intellectual property rights, the project delivers an innovative solution for intelligent prosthetics, helping users regain daily and working abilities with advanced human-machine interaction.



Post-processing Algorithm for One-to-Many Mapping

2. Overall Architecture

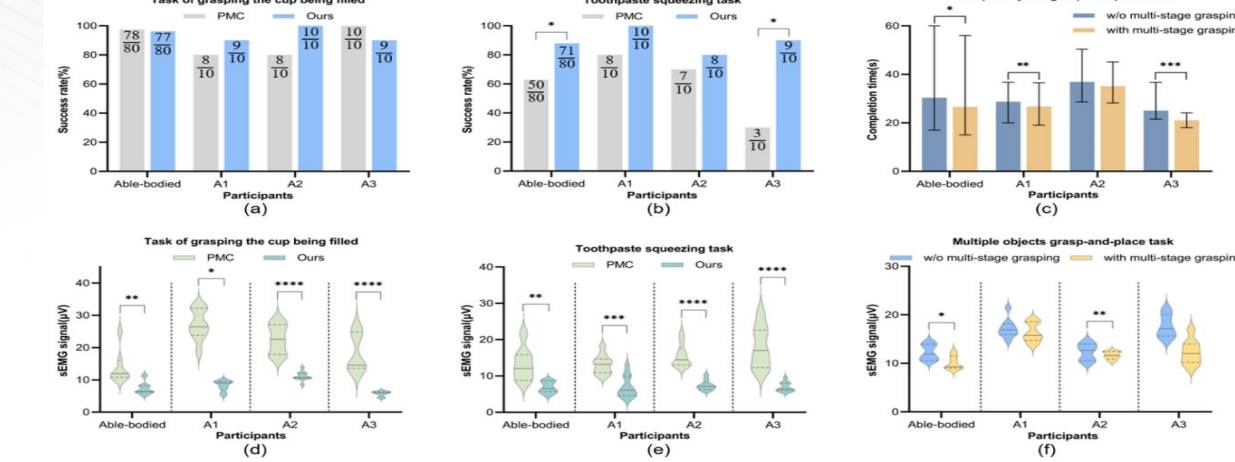
This diagram illustrates the five-layer closed-loop architecture of the Zhiyu Lingzhi prosthetic system, inspired by the brain's "sense-perceive-control-learn" logic:

- 1.Perception:** Bionic hardware + multi-modal sensors collect user and environmental data.
- 2.Decoding:** The DiffSR model processes and interprets myoelectric signals.
- 3.Control:** Human-machine shared control and TKE-BGC skill transfer enable complex tasks.
- 4.Execution:** High-precision actuators with personalized calibration.
- 5.Evolution:** Continuous optimization via data recording and skill enhancement.

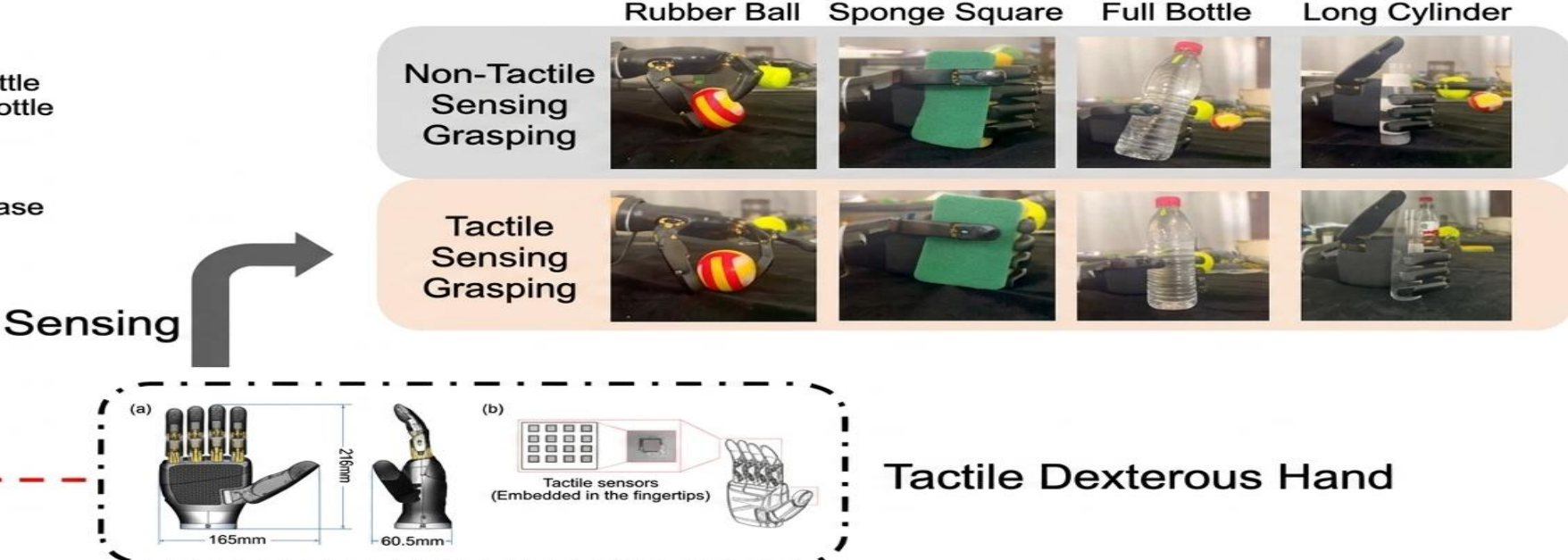
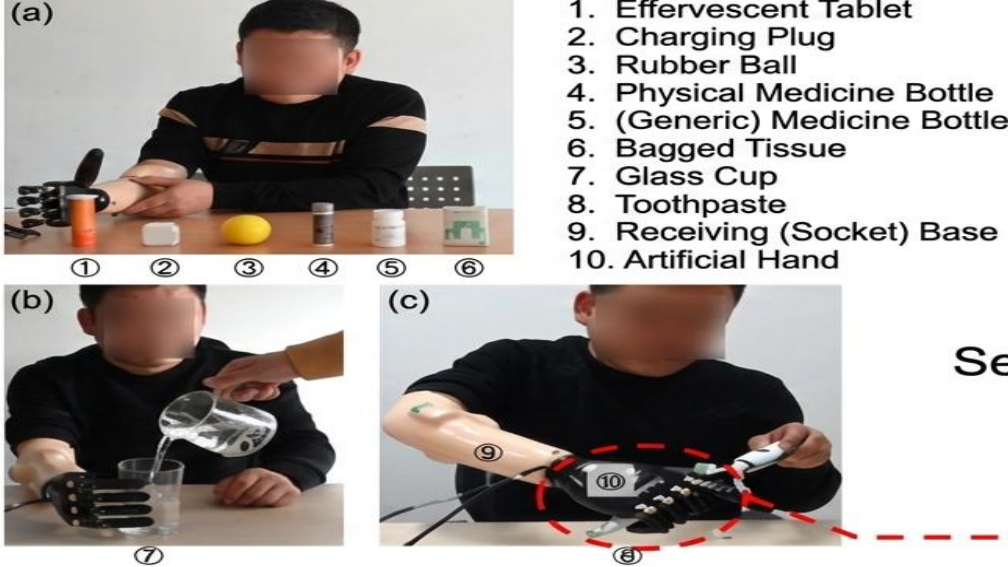


3. Tech1:Full-Dimensional Tactile Perception & Precise Intention Recognition

- High-density sEMG sensor array + high-resolution flexible tactile sensor
- Autonomous grip force adjustment based on real-time contact feedback
- Task success rate: **70%–85%**; Task time reduced by **26.5%**
- Solves: Lack of tactile sensing, rigid grasping, low fault tolerance

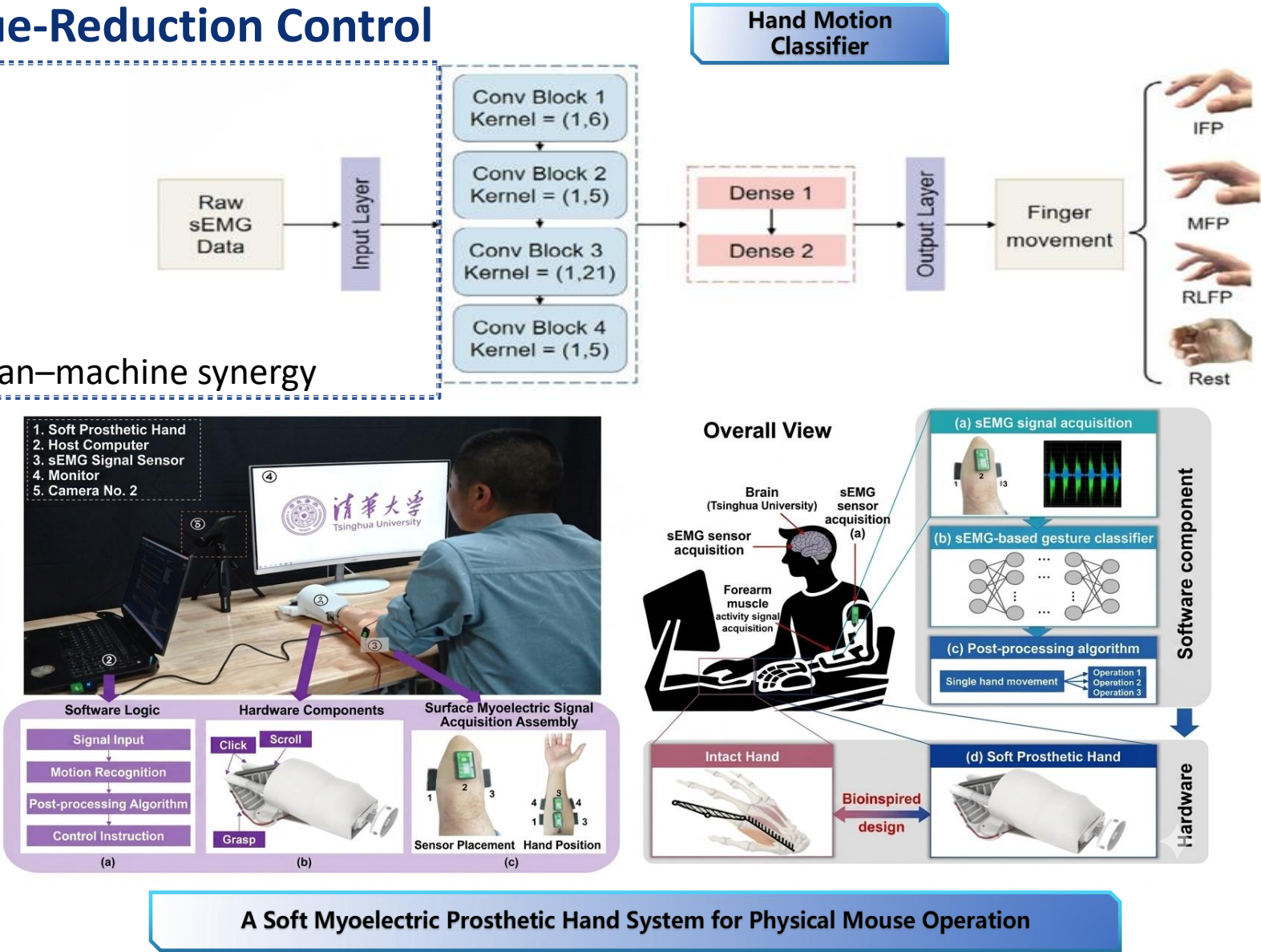
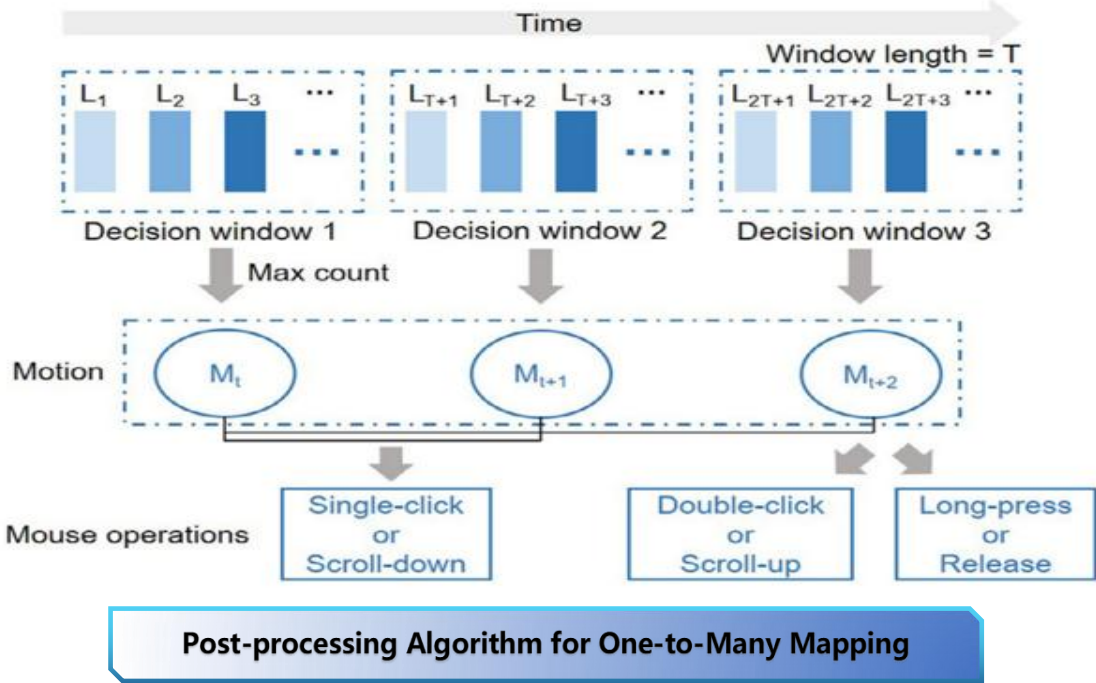


A Tactile-Based Shared Control Scheme for Dexterous Prosthetic Hands:



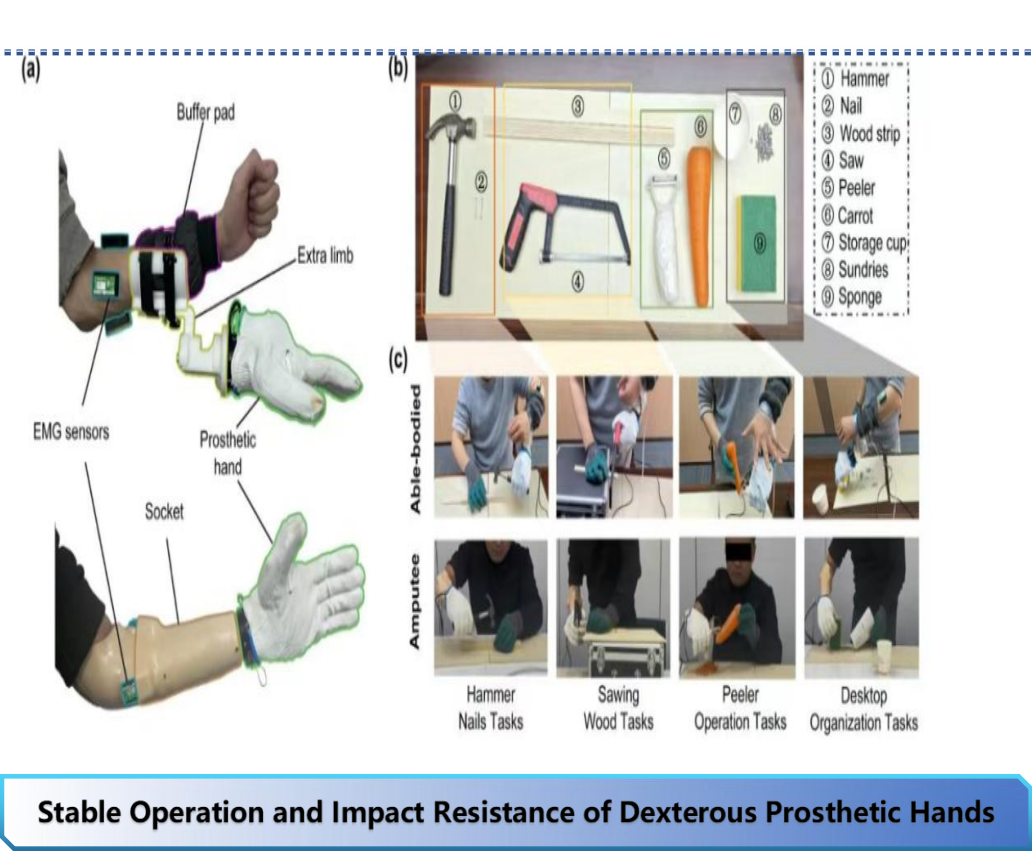
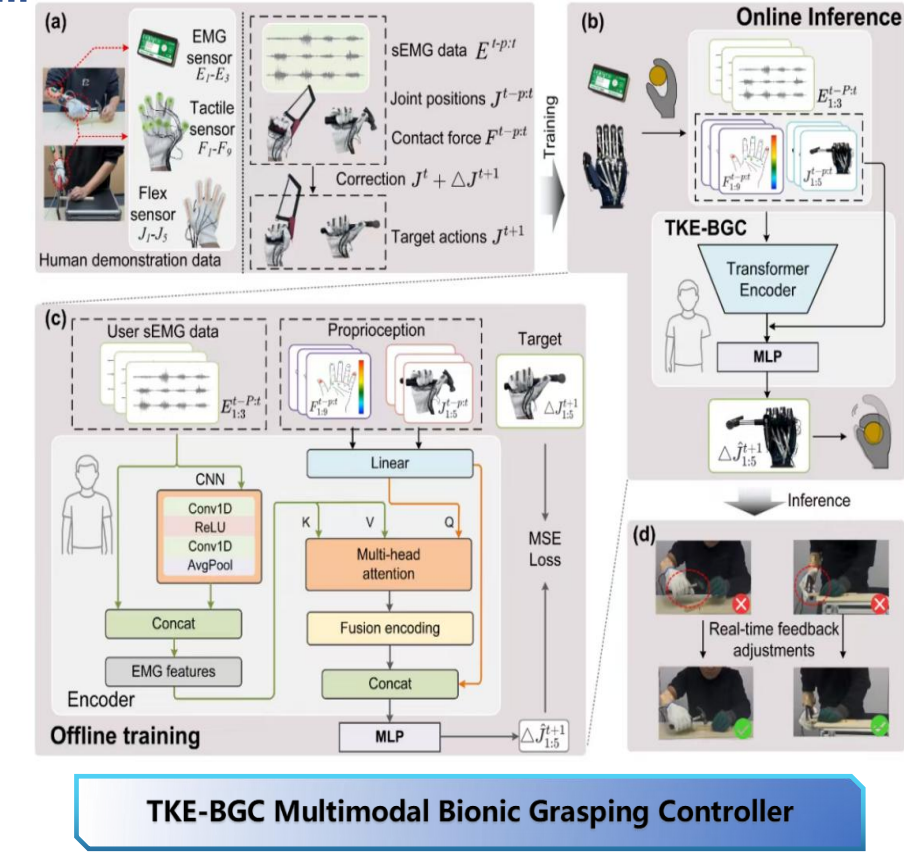
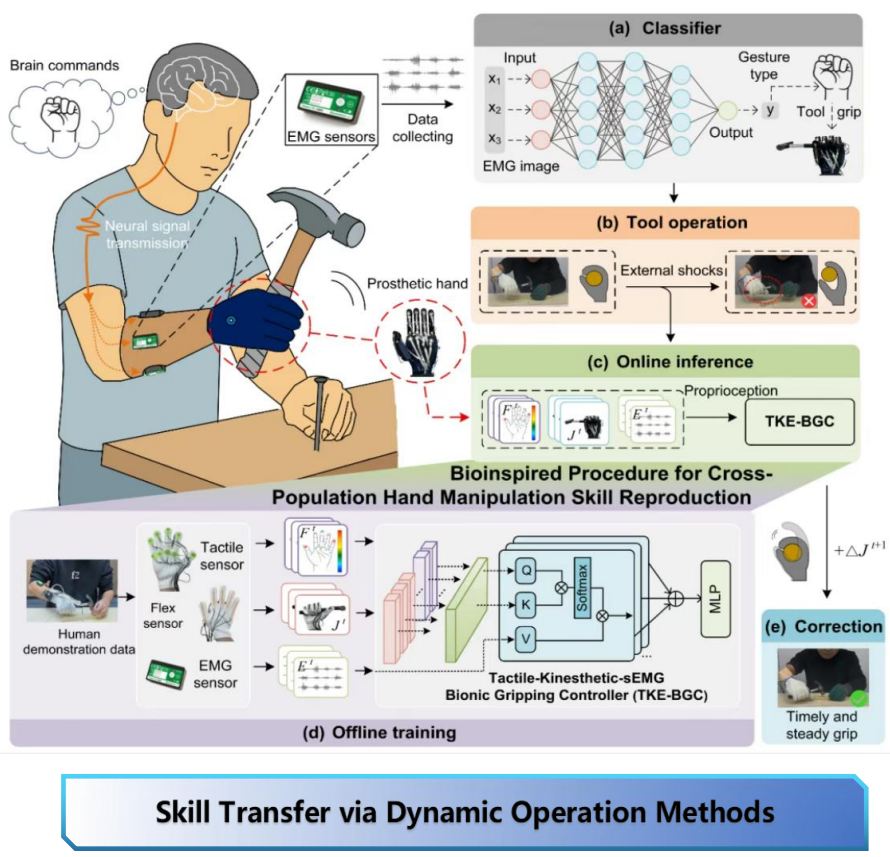
4. Tech2:Human-Machine Collaboration & Fatigue-Reduction Control

- Tactile-intelligence-driven shared control strategy
- One-to-many mapping** algorithm lowers training burden
- Significantly reduced sEMG activation & muscle fatigue
- Solves: Continuous muscle tension, mental fatigue, poor human-machine synergy



5. Tech3:Skill Transfer & Complex Dynamic Manipulation

- TKE-BGC multi-modal bionic controller
- Enables dynamic tasks: twisting, tapping, cutting, tool operation
- Human-like force control, high stability & shock resistance
- Solves: Only static grasping, limited employment, narrow scene adaptability



Contact:

This is the wechat QR code of the leader of our project, if you have any questions, please do contact.

